

Abstract:

This project seeks to identify axes by which inequality of educational opportunities arises in U.S. high schools. As a proxy for student achievement, we need a standardized measure, so we chose to use average performance on ACT or SAT exams, which are typically taken as part of the process of applying for college. In addition to factors such as median income for the census tract or percentage of married caregivers, I was interested to know if student/teacher ratios would impact student achievement. Before high school, I attended a private school known for its low student-to-teacher ratio. I'm curious if this access to faculty made a difference. To answer this question, we gathered data from EdGap and the National Center for Education Statistics. We then cleaned the data using best practices and prepared it for analysis. We used linear regression models to measure how socioeconomic factors relate to SAT scores. While the variables all had some impact on SAT scores, access to free and reduced lunch was the strongest predictor of scores and allowed us to calculate SAT scores to within one point on average. This provides a rich ground—beyond the scope of this paper—to consider why free and reduced lunch strongly negatively correlates with SAT scores. One should also consider that most of the other socioeconomic factors were measured in the general community nearby; i.e., rich childless individuals might skew median income results, whereas lunch was measured at the school itself.

Introduction:

We want to see how socioeconomic factors affect student performance in school. In order to do so, we need some sort of standardized method of comparison, so we chose to use SAT or ACT scores as a proxy for student achievement. These are national standardized tests that many seniors in high school take as part of the college admissions process. However, not every high school requires its students to take these tests, and not every high school reports its scores. Considering these points, we will proceed, mindful of these challenges as we reach conclusions.

One data source for this analysis is EdGap.org¹. This organization provides insights into “opportunity gaps” affected by socioeconomic factors. EdGap determines socioeconomic data (household income, unemployment, adult educational attainment, and family structure) from the Census Bureau’s American Community Survey. Helpfully, EdGap has converted SAT scores to equivalent ACT scores and associated all information with the National Center for Education Statistics school identification number or NCESSCH. It is also important to keep in mind that school districts can pull from multiple census tracts. For 2016, our year of interest, EdGap only has information for 20 schools, which is another consideration to keep in mind while interacting with this data. This data was already in a clean and tidy format when loaded and only contains columns that are of interest to us: NCESSCH School ID, CT Unemployment Rate, CT Pct Adults with College Degree, CT Pct Children in Married Couple Family, CT Median Household Income, School ACT average (or equivalent if SAT score), and School Pct Free and Reduced Lunch. In this context, CT means “census tract,” pct is percentage, and ACT scores are out of 36 possible.

¹ “About the EDGAP Map: Visualizing the Education Gap.” *EdGap.Org | SAT/ACT College Readiness Map*, www.edgap.org/. Accessed 22 Oct. 2025.

The secondary data source used in this analysis is from the National Center for Education Statistics and contains basic information about the schools to supplement our data from EdGap. We will use this dataset to ensure that the schools we are referencing from EdGap are high schools and see if there is any difference between different types of schools, such as charter schools. This dataset is more extensive since it reports every school in the U.S., and we are only interested in the following columns: SCHOOL_YEAR, STATENAME, SCH_NAME, NCESSCH, LSTATE, LZIP, SCH_TYPE_TEXT, LEVEL, CHARTER_TEXT. LEVEL denotes whether the school is an elementary school or high school, and CHARTER_TEXT indicates whether it is a charter school or not.

The tertiary data source was from the National Center for Education Statistics as well and adds information about the student-to-teacher ratio at each school. From this dataset, we are primarily interested in the following columns: School ID (12-digit) - NCES Assigned [Public School] Latest available year, Pupil/Teacher Ratio [Public School] 2016-17. There are additional columns that provide location data, which will mostly be used to ensure that the information matches up with the schools from the other data.

Thankfully, the datasets are already in the tidy format. Additionally, both datasets feature a column for the NCESSCH, or school identifier number, which will be a convenient way to merge the two dataframes and ensure we are keeping the information consistent across schools. Unfortunately, the NCESSCH will need to be standardized, since it is stored as different data types in each dataset (object in EdGap and float64 in the school information dataset). Simply deleting the rows with NaN or missing values will negatively impact the socioeconomic factors, so we will impute the missing values using the iterative imputer tool from Scikit-learn. This process will be explained in more detail in the methodology section. Overall, iterative imputation replaces missing values with successive adjustments of the mean, evaluated in comparison to the effect of the other variables and known values.

Methodology

The tools we used to initially refine our data, and then explore it, are pairplots, imputation, a correlation matrix, and linear regression.

Through the use of pairplots, we were able to determine if our data would be able to address our question, identify the presence of outliers, and observe relationships between the socioeconomic factors and ACT scores. Since the pairplots each had a structure in the scatterplots, we were able to conclude that there is a relationship between these socioeconomic factors and ACT scores, and overlaid regression lines give a rough idea of the strength and direction of the correlation. The scatterplots also depict some out-of-range ACT, percent lunch values, and student-teacher ratios, which were dealt with by turning them into NaN, or missing values. While working with the data, it is also important to notice that we do not have data for every state. This is not by virtue of missing values but from the collection method. Not every school has reported its information to EdGap, so we are only seeing a subset of schools from twenty states.



Fig 1. *Fig 1. A map of the US with outlines of each state. States are filled in on a gradient based on how many schools reported their ACT/SAT information to EdGap. There are 20 states depicted.*

Next, imputation was used to replace missing values, since simply deleting these values would adversely affect our analysis. More about how the scikit-learn imputer works can be found on the documentation page², but in brief, the process is as follows. The algorithm replaces the missing value with a value based on the mean while using information from the other variables. The estimated value is evaluated in comparison to known observed values and adjusted based on how well it performs. The process continues across all variables and missing data until there are no more missing values.

Using our clean data, we constructed a correlation matrix to quantitatively observe relationships between the data. Already, we can identify some variables that might have stronger impacts on ACT scores, though they all seem to contribute in some way, except for student-teacher ratio, since none of the other correlations are less than 43% correlated to ACT scores. We can also see that some of the scores might be related to each other, such as median_income and percent_college, which are 71% correlated.

² "Iterativeimputer." Scikit, scikit-learn.org/stable/modules/generated/sklearn.impute.IterativeImputer.html. Accessed 22 Oct. 2025.

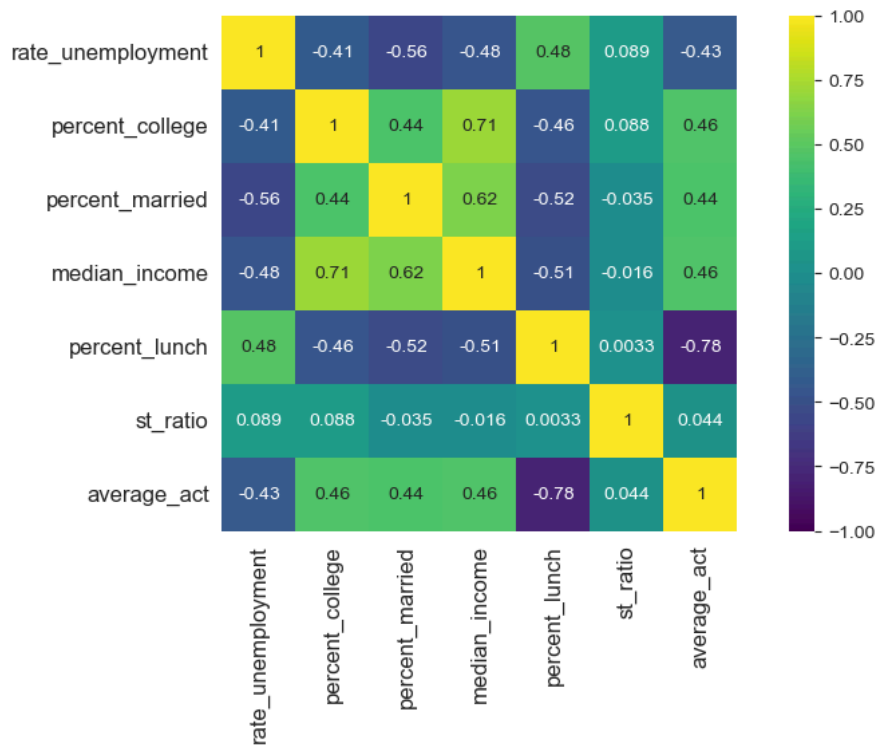


Fig 2. A correlation matrix of the socioeconomic factors and average ACT score. The highest correlation is percent lunch, while the lowest is the student-teacher ratio. We can also see that median income and percent college are strongly related to each other, so we might not need both of them for analysis.

Finally, we applied linear regressions to determine which socioeconomic factors can best be used to predict SAT scores. Via the scatterplot, we were able to observe a linear relationship between the data, and we can be fairly confident that the factors are independent of each other. One can approach setting up linear regression in a few ways, but we chose to start with single linear regressions looking at the contribution of each variable to SAT scores. Then we made a mult-linear regression model with all of the factors equally weighted to understand how they relate to each other. Based on the multi-linear regression model, we determined that not all of the factors contributed as much to the score, so we were able to build a reduced model with only the most significant contributions. Finally, we scaled each factor. Since the socioeconomic factors are all based on different units in different scales (for example, \$100,000 for median income vs. 1.5 for percent lunch), we transformed the data so that one unit change in one variable would be equivalent to one unit change for the others. This allows us to directly compare the coefficients of our model and make conclusions.

Computational Results:

	coef	std err	t	P> t	[0.025	0.975]
Intercept	22.4093	0.146	153.525	0.000	22.123	22.695
rate_unemployment	-2.5240	0.406	-6.224	0.000	-3.319	-1.729
percent_college	1.5988	0.159	10.057	0.000	1.287	1.910
percent_married	-0.0813	0.134	-0.609	0.542	-0.343	0.180
median_income	4.062e-07	1.21e-06	0.336	0.737	-1.97e-06	2.78e-06
percent_lunch	-7.5971	0.097	-78.636	0.000	-7.786	-7.408
st_ratio	0.0225	0.004	5.807	0.000	0.015	0.030

Fig3. Shows the outputs of the multilinear regression model and specifically the coefficient summary. From this, we can see that percent married, median income, and student-teacher ratio do not contribute much to the model since their coefficients are small.

```
print(model_normalized.summary())
```

OLS Regression Results						
Dep. Variable:	average_act	R-squared:	0.628			
Model:	OLS	Adj. R-squared:	0.628			
Method:	Least Squares	F-statistic:	4064.			
Date:	Wed, 22 Oct 2025	Prob (F-statistic):	0.00			
Time:	19:28:09	Log-Likelihood:	-13328.			
No. Observations:	7227	AIC:	2.666e+04			
Df Residuals:	7223	BIC:	2.669e+04			
Df Model:	3					
Covariance Type:	nonrobust					
	coef	std err	t	P> t	[0.025	0.975]
Intercept	20.2986	0.018	1127.597	0.000	20.263	20.334
rate_unemployment_normalized	-0.1227	0.021	-5.804	0.000	-0.164	-0.081
percent_college_normalized	0.2826	0.021	13.502	0.000	0.242	0.324
percent_lunch_normalized	-1.7770	0.022	-81.995	0.000	-1.819	-1.734
Omnibus:	873.069	Durbin-Watson:	1.483			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	3128.272			
Skew:	0.587	Prob(JB):	0.00			
Kurtosis:	6.002	Cond. No.	1.93			

Fig 4. Shows the results of linear regression using the reduced and standardized model with unemployment rate, percent college, and percent of free lunch included. Percent lunch has the largest coefficient, indicating the largest contribution to the prediction.

Once scores were standardized, the coefficient for rate of unemployment is -0.122, percent_college is 0.28, and percent_lunch is -1.77. Therefore, we are able to determine that, of the factors we tested, percent_lunch is the single best predictor of ACT scores.

Discussion:

Percent school lunch at a school is a very strong predictor of ACT scores. Importantly, school lunch is a metric that was measured at the school, whereas other metrics are based on census tracts and are not as reflective of the lived experience at the school. I am surprised by the

minimal impact of student-to-teacher ratio and believe this is worth exploring more. Overall, there are many factors that contribute to educational opportunity, and it is interesting to ponder why access to free lunch might be so strongly associated with lower ACT scores.

References:

“About the EDGAP Map: Visualizing the Education Gap.” EdGap.Org | SAT/ACT College Readiness Map, www.edgap.org/. Accessed 22 Oct. 2025.

“Iterativeimputer.” Scikit, scikit-learn.org/stable/modules/generated/sklearn.impute.IterativeImputer.html. Accessed 22 Oct. 2025.